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Diesel Hilux GEET conversion instructions

- All information and credits belong to Ady and Naresh

Background by Naresh

If you want the reactor or reactors could be for water only. In 1 way that is easier but harder at the same time. What I mean is, it is harder to crack plain water but even if the reactor fails to crack the water, water vapours in the cylinders may give an improvement in fuel economy. If battery fumes or an electrolysis cell fumes or both are mixed with the water vapours then it may be easier to crack them in the reactor.

The reactor fuel of choice affects the tube in tube length. If it's going to be the vapours of water and electrolytes then many short reactors are best. If the reactor fuel is going to be water and electrolytes and well atomized diesel fuel then few and longer reactors are needed. Then the next big design task is to determine the best way to keep a good vacuum in the reactors and also keeping the output plumbing short , smooth and probably made of stainless spiral tubing or copper tubing.

If the engine has glow plugs you will likely want to replace them with non-resistive spark plugs timed to fire a plasma boost circuit into each spark plug at the timing when the fuel is injected, rather than before or after fuel injection. But the ignition timing should still be made easily adjustable for to test what gives the best result. An electronic circuit with a potentiometer to vary the timing would be good.

There are a lot of ways to do that. A crank sensor at 90 degrees BTC could trigger a variable delay circuit and the pot adjusted from 90 degrees BTC to 90 degrees ATC. The original diesel injectors should keep injecting for cylinder's lubrication I think. I'm guessing that size tube and rod needed to fit in that 1 cylinder's straight horizontal section of exhaust tubing will be

Something like:

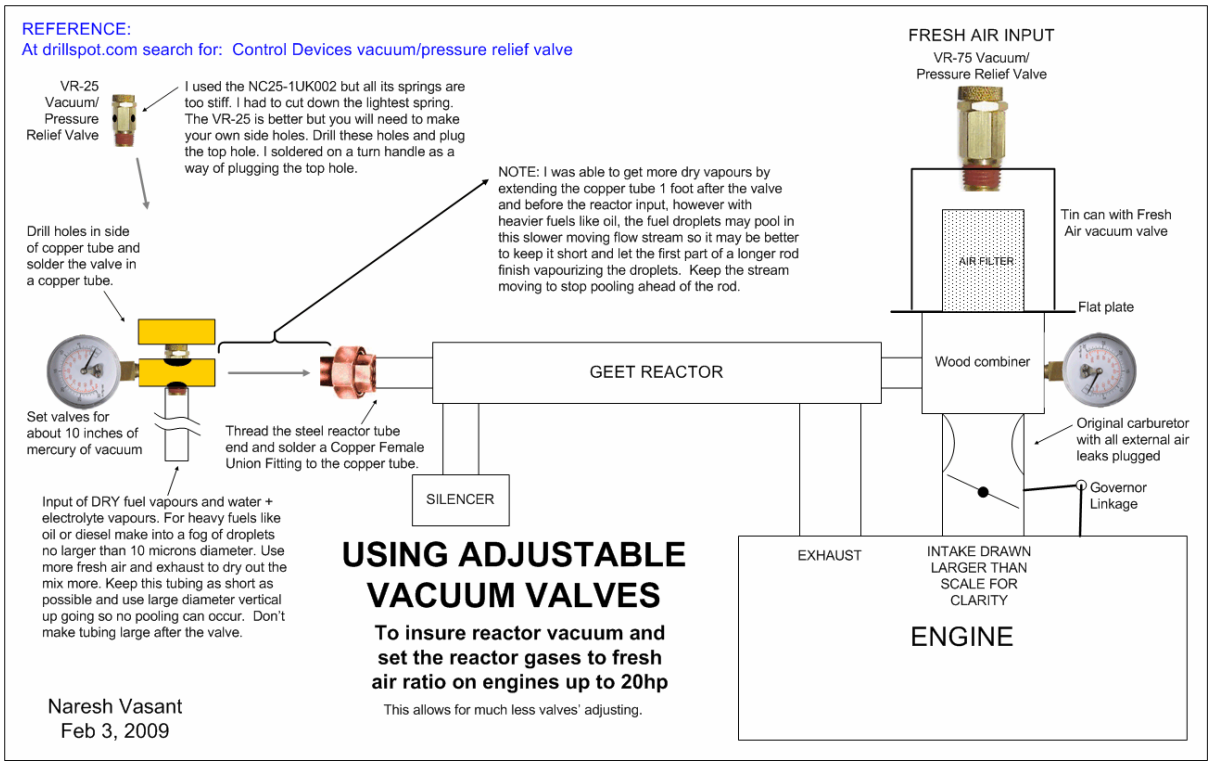
Rod: cold or hot rolled 3/8 inch diameter or about 9.5mm. Tube: steel tube with a smooth inner diameter of about 10.3mm and a wall thickness anywhere from 1.5mm to 2.0mm.

It needs to be installed in a way that allows access to take the rod in and out. There might be a getting to the rear end to push the rod out if it gets stuck. It looks like there is not even much room to put a right angle fitting close to the firewall.

The rod length can be about 11 inches long if possible. It seems like that straight section of exhaust pipe may be a little short for diesel as reactor fuel. If a bubbler is used with just a little diesel fuel and a lot of water + electrolyte then if that tube in tube section ends up being too short then maybe the rod can be tuned for water as fuel.

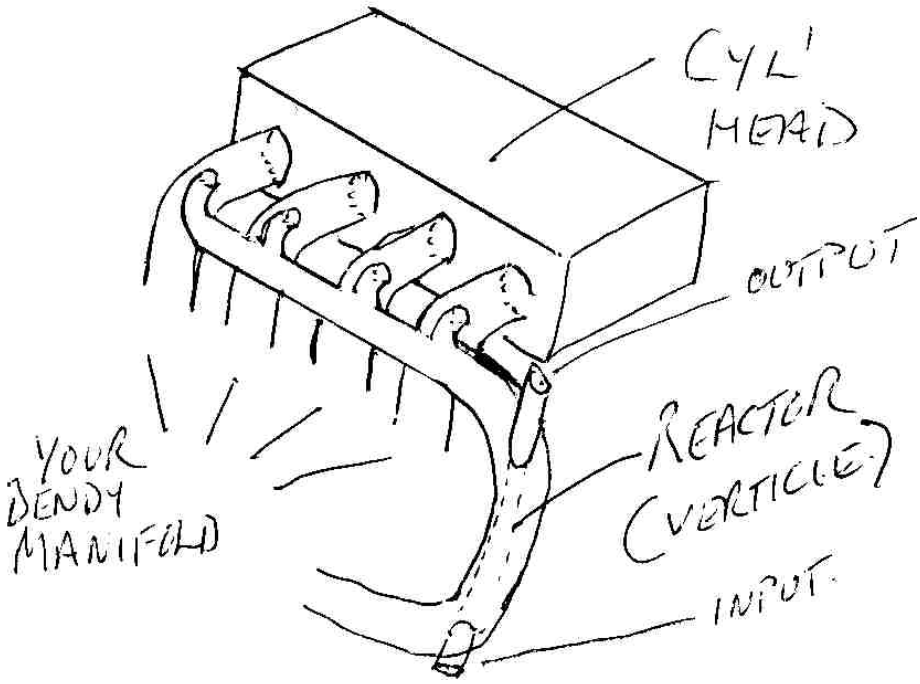
We can just put it all together with de-magnetized parts and see how long of a magnetic signature develops. I suggest using a large and a small vacuum valve like in this diagram:

The valve positions would be similar used on this petrol engine, like this:



Except you will not have the carburettor and its linkage so the combined gases would just go straight into the engine intake. The main thing is to keep at least 10 inches of vacuum going to the reactor. The larger fresh air valve can be made from a flap and a spring or bungy cord. Else, just block off a fixed amount of the air such that at medium to high RPM the vacuum tends to be at least 10 inches or more.

Ady has a good idea. Just leave the existing exhaust manifold rather than replace it, then make holes in the individual exhaust output tubes and connect a new exhaust manifold specifically for the reactor. I suggest using thin wall steel tubing that can be welded but not too heavy so the weight of it will not pull too much on the old exhaust manifold.



Construction details:

Just some questions on the following diagram (questions are drawn into the diagram) as I can't quite make sense of it:

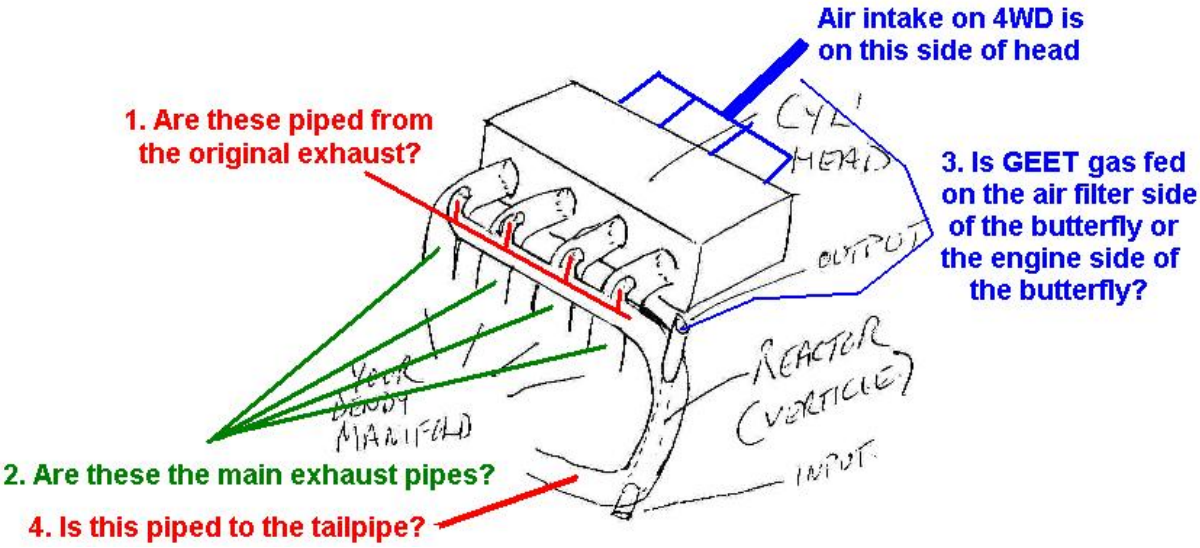
1. Are these piped from the original exhaust?

2. Are these the main exhaust pipes?

3. Is the GEET gas fed on the air filter side of the butterfly or the engine side of the butterfly

4. Is this piped to the tailpipe?

If you could elaborate / expand on the drawing (show some more of the manifold piping) that would make it easier to understand Further R and D down the Track.



Answers by Naresh and approved by Ady

1. Yes, holes are made to that original exhaust manifold where a new exhaust manifold will be welded on.

2. Yes, those are the pipes of the original exhaust manifold of the truck.

3. GEET gas enters on the air filter side of the throttle plate, between the throttle plate and the air filter. Cover the input of air to the filter box with something to cause the vacuum going to the reactor to be at least 10 inches of vacuum at the reactor's inner tube input side going to the bubbler. A spring loaded flap or spring loaded ball against a hole or something can open more as the engine rev.s up and sucks more air but it must never let so much air through that the vacuum going to the reactor ever goes less than 10 inches, even a little more might be better.

If it works better than expected then you will have a problem that you cannot turn off the engine because even if the truck is off and not injecting diesel, the engine could possibly still keep running on GEET gases. In that case, to stop the reactor from outputting GEET gas, you will need to add a different kind of vacuum relief valve that works just the opposite, meaning if you pull a cable to open it then the reactor will lose its vacuum and that stops the reactor from working so the engine can die. That cut off valve needs to be bigger so there is a definite loose of vacuum to the reactor when that valve is opened.

4. Yes, but you also want to Tee off that pipe to send the exhaust gases into a bubbler container with water + electrolyte + diesel fuel in it.

Or you may want to just try it with only water + electrolyte in the container. But sending the exhaust gases to that bubbler tends to keep it full with water because so much water will condense out of the exhaust gases and tend to keep the level from going down in the container.

Between that bubbler container's output and the reactor inner tube's input place a vacuum relief valve that get sucked open anytime the vacuum through the reactor inner tube is at least 10 inches of mercury.

It can sometime help to let a small amount of fresh air bleed in and mix with the bubbler output so both go together into the reactor but that fresh air needs to come in through its own vacuum valve as well.

The rod should not need centring posts and the inner tube should be very smooth inside to help the rod to start spinning. If the rod gets sucked up to the top, if should have a rear stop that sets into the rod's rear dimple to allow the rod to still spin with the rear stop as a pivot point.

Further R and D

This concept was suggested by Ady and refers to the science used in the Sebatier effect. It's to do with using Nickel as a catalyst and is what has given Ady the idea of getting the Reactor Rods Electrodes Nickel plated. Ady suggests getting the Rod platted for the Hylux, and swears that it will make a whole lot of difference in cracking water, on my life.

Technical support

http://groups.yahoo.com/group/VortexHeatExchanger/

Credits

The open source energy community

If you are able to contribute to this document in ANY way, IE- replication details, faculty info and or additional data please contact the non profit organization.

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